The comparative method

Quantitative Methods in Historical Linguistics

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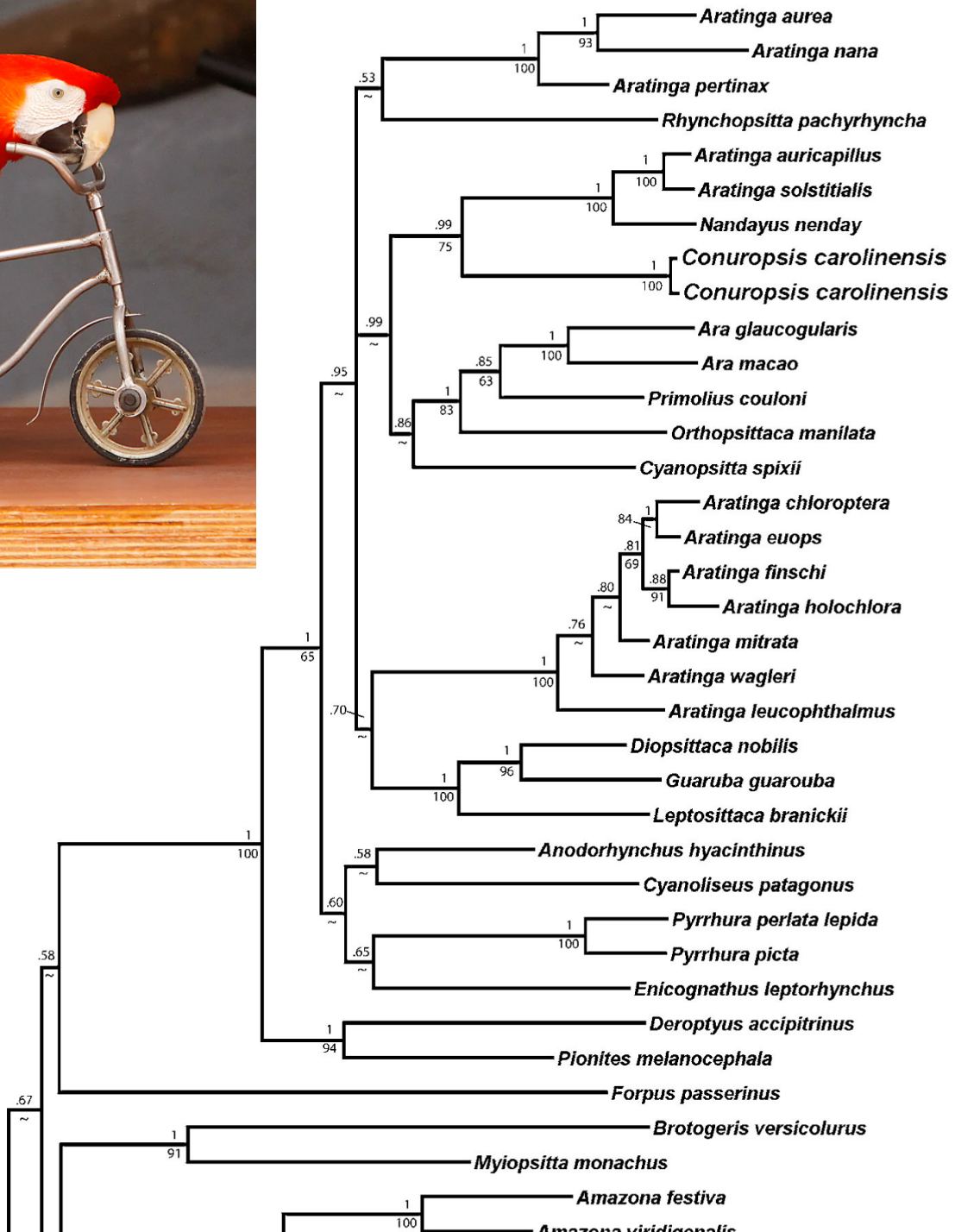
First things first

- Questions from last time?
- How are you getting along with R?



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and
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IPA drill

- A lot of what follows turns on our ability to theorize about sound changes

⇒ Need a phonetic alphabet

- We will use the International Phonetic Alphabet (IPA)

Exercise

Transcribe the following English and German words into IPA:

1. shoe
2. socks
3. trousers
4. pullover
5. stockings
6. Raum
7. Seminar
8. Veranstaltung
9. lernen
10. schreiben

Language families

- Languages are sorted into **LANGUAGE FAMILIES** based on how closely related they are
- E.g. German, Dutch, English → Germanic family
- French, Spanish, Italian → Romance family
- Three important questions:
  1. How are language families established?
  2. What does it mean for two languages to be ‘related’?
  3. How is linguistic knowledge transmitted over time?

How is linguistic knowledge transmitted?

- Every healthy human child acquires a first language (Muttersprache), usually abbreviated **L1**, if subjected to sufficient linguistic input during a critical period
  - The linguistic input comes from the parents' speech, from relatives, friends and so on… briefly put, from the child's **LINGUISTIC ENVIRONMENT**
  - Critical period: a period of time during which the brain is especially good at acquiring an L1 (after this, it will be difficult/impossible)
- This **L1 ACQUISITION** is the main process whereby linguistic knowledge is transmitted from one generation of speakers to the next (cp. inheritance in biology)

How is linguistic knowledge transmitted?

- Example: I acquired Finnish, my L1, (mostly) from my parents, who acquired it from their parents, who…
- But if we go back sufficiently far in this chain, we start to see the language changing!
- This is because L1 acquisition isn't always reliable (cp. mutations in biology)
- Languages change in many ways
  1. Phonetic change
  2. Phonological change
  3. Morphosyntactic change
  4. Semantic change
  5. Lexical change

Reflection Exercise

Can you think of one example of each of 1–5?

When are two languages related?

- Language A is said to be a **DESCENDANT** of language B (or to descend from B) if there is a chain of L1 acquisition events from B to A
- We also say that B is an **ANCESTOR** of A
- Examples:
  - Modern English is a descendant of Old English
  - Modern Spanish is a descendant of some form of Latin
  - Modern Finnish descends from something known as Proto-Uralic
  - The Finnish I speak (as an L1) is a descendant of the Finnish my parents speak
- Non-examples:
  - Modern English does not descend from Latin
  - Modern Spanish does not descend from Modern Italian

When are two languages related?

- Two languages A and B are said to be **RELATED** if they share a common ancestor
  - NB: if language evolved only once (the **MONOGENESIS** hypothesis), then all languages are trivially related. But this is not usually the intended meaning.
- A possibly better strategy is to talk about the **DEGREE OF RELATEDNESS** of two languages
- Intuitively, Spanish and Portuguese are closer related than Spanish and German, for example
- Main question during the first part of this course: how can we make this notion of 'degree of relatedness' precise?

How are language families established?

- A language family is a group of closely related languages
- How can we demonstrate that two languages are closely related enough to belong to the same family?
- Classically, one uses the comparative method (CM)
- The CM is a method for the **RECONSTRUCTION** of ancestor languages
- Indirectly, it also gives information about language relatedness
- The method relies on the detection of **SYSTEMATIC SOUND CORRESPONDENCES**
- Time-consuming to do; a lot of 'philological detective work' is involved

CM, step 1: compile cognate sets

- The first step in the CM: pick a set of languages and words which we think might be related
- (Words which descend from a common ancestor word are known as **COGNATES**)
- Put these words in a table and compare the phonological forms of the words across the languages

Italian	**Spanish**	**French**	**gloss**
capra /kapra/	*cabra* /kabɾa/	*chèvre* /ʃɛvʁ/	'goat'
caro /karo/	*caro* /karo/	*cher* /ʃɛʁ/	'dear'
capo /kapo/	*cabo* /kabo/	*chef* /ʃɛf/	'head, top'
cane /kane/	*can* /kan/	*chien* /ʃjɛ̃/	'dog'

CM, step 2: detect sound correspondences

Italian	**Spanish**	**French**	**gloss**
capra /kapra/	*cabra* /kabɾa/	*chèvre* /ʃɛvʁ/	‘goat’
caro /karo/	*caro* /karo/	*cher* /ʃɛʁ/	‘dear’
capo /kapo/	*cabo* /kabo/	*chef* /ʃɛf/	‘head, top’
cane /kane/	*can* /kan/	*chien* /ʃjɛ̃/	‘dog’

- Next we detect **SOUND CORRESPONDENCES**, e.g.:

#k : #k : #ʃ

- Read this: where Italian and Spanish have /k/ word-initially, French has /ʃ/
- (I use # to mark word boundaries)

CM, step 2: detect sound correspondences

Italian	**Spanish**	**French**	**gloss**
capra /kapra/	*cabra* /kabɾa/	*chèvre* /ʃɛvʁ/	‘goat’
caro /karo/	*caro* /karo/	*cher* /ʃɛʁ/	‘dear’
capo /kapo/	*cabo* /kabo/	*chef* /ʃɛf/	‘head, top’
cane /kane/	*can* /kan/	*chien* /ʃjɛ̃/	‘dog’

- Another sound correspondence:

V# : V# : ∅#

- Read this: where Italian and Spanish have a vowel at the end of a word, French has nothing
- (I use V to stand for any vowel; ∅ means that something has been deleted)

CM, step 2: detect sound correspondences

Italian	**Spanish**	**French**	**gloss**
capra /kapra/	*cabra* /kabɾa/	*chèvre* /ʃɛvʁ/	‘goat’
caro /karo/	*caro* /karo/	*cher* /ʃɛʁ/	‘dear’
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- Yet another:

  VpR : VbR : VvR

- Read this: where Italian has /p/ between a vowel and a rhotic, Spanish has /b/ and French has /v/
- (So R stands for rhotics (“R-sounds”), sounds like /r/ (alveolar trill), /ɾ/ (tap) and /ʁ/ (uvular trill))

CM, step 3: reconstruct proto-phonemes

- The next step is to reconstruct what the most likely phonemes of the ancestor language were
- This is based on the sound correspondences we have detected
- We are guided by two principles:
  - **Majority wins**: reconstruct that phoneme which is most frequent among the daughter languages
  - **Directionality**: if there is a choice to reconstruct either of the two pathways A > B and B > A, where A and B are two phonemes, reconstruct the one that is more likely on independent grounds

Principle 1: majority wins

- Recall our first sound correspondence:

#k : #k : #ʃ

- Now, we have three options:
  1. Reconstruct #k for Proto-Romance
  2. Reconstruct #ʃ for Proto-Romance
  3. Reconstruct something else for Proto-Romance
- The consequences of these reconstructions are:
  1. One language has changed
  2. Two languages have changed
  3. Three languages have changed

⇒ Choose reconstruction 1

Principle 2: directionality

- Recall our third sound correspondence:

VpR : VbR : VvR

- In this case, the majority wins principle does not give a clear-cut winner. What to do?
- Changes like p > b and b > v are common when the consonant occurs between two sonorants (such as a vowel and a rhotic)
- This is known as **LENITION**, and it has a well-understood physiological basis
- Changes like b > p and v > b are uncommon

⇒ Reconstruct VpR and assume lenition has occurred

CM, step 4: resolve overlapping correspondences

- Let's add four more cognate sets to our data:

Italian	**Spanish**	**French**	**gloss**
capra /kapra/	*cabra* /kabɾa/	*chèvre* /ʃɛvʁ/	'goat'
caro /karo/	*caro* /karo/	*cher* /ʃɛʁ/	'dear'
capo /kapo/	*cabo* /kabo/	*chef* /ʃɛf/	'head, top'
cane /kane/	*can* /kan/	*chien* /ʃjɛ̃/	'dog'
colore /kolore/	*color* /kolor/	*couleur* /kulœʁ/	'colour'
correre /korere/	*correr* /korer/	*courir* /kuʁiʁ/	'to run'
costare /kostare/	*costar* /kostar/	*coûter* /kute/	'to cost'
cura /kura/	*cura* /kura/	*cure* /kyʁ/	'cure'

- Rows 1–4 suggest the correspondence #k : #k : #ʃ
- Rows 5–8 suggest the correspondence #k : #k : #k

CM, step 4: resolve overlapping correspondences

- So we need to reconcile #k : #k : #ʃ and #k : #k : #k
- Three options:
  1. The protolanguage had both #k and #ʃ. French maintained this distinction, but Italian and Spanish **MERGED** the two phonemes
  2. The protolanguage had #k, which French **SPLIT** into #k and #ʃ
  3. The protolanguage had #ʃ, which Italian and Spanish split into #k and #ʃ
- Both majority wins and directionality support reconstruction 2.
- Let's have another look at the data…

CM, step 4: resolve overlapping correspondences

Italian	**Spanish**	**French**	**gloss**
capra /kapra/	*cabra* /kabɾa/	*chèvre* /ʃɛvʁ/	‘goat’
caro /karo/	*caro* /karo/	*cher* /ʃɛʁ/	‘dear’
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cane /kane/	*can* /kan/	*chien* /ʃjɛ̃/	‘dog’
colore /kolore/	*color* /kolor/	*couleur* /kulœʁ/	‘colour’
correre /korere/	*correr* /korer/	*courir* /kuʁiʁ/	‘to run’
costare /kostare/	*costar* /kostar/	*coûter* /kute/	‘to cost’
cura /kura/	*cura* /kura/	*cure* /kyʁ/	‘cure’

- A plausible story is:
  1. French first turned a into ɛ…
  2. …and then underwent the change kɛ > ʃɛ
     (when followed by vowels other than ɛ, k remained)

CM, step 5: sanity checks

- With these tools, *and using much, much more data*, one can try and reconstruct the entire phoneme inventory and phonotactics of the protolanguage
- The next step is to see if the resulting system “makes sense” from a synchronic or typological viewpoint
- “Suspicious” reconstructed phoneme inventories:
  - /ɑ̃/ /õ/ /ɛ̃/ /ĩ/
  - /i/ /e/ /a/
- Suspicious reconstructed sequences:
  - /pgtkb/
  - /baeioup/
- And so on

CM and language relatedness

- The fact that the CM can be successfully applied to Italian, Spanish and French to reconstruct a protolanguage means that the three languages are related
- It worked because the sound correspondences we found were **SYSTEMATIC**
- E.g. for (virtually) any word of Italian which has VpR, it is true that the cognate Spanish word has VbR

Neogrammarian regularity hypothesis

Sound change is regular in the sense that it applies across the lexicon without exceptions.

- Let's see what happens when the correspondences are **NOT** systematic…

False cognates

- English has *day* /deɪ/, Spanish has *día* /dia/ — are English and Spanish related?

Italian	**Spanish**	**English**
dì /di/	*día* /dia/	*day* /deɪ/

False cognates

- English has *day* /deɪ/, Spanish has *día* /dia/ — are English and Spanish related?

Italian	**Spanish**	**English**
dì /di/	*día* /dia/	*day* /deɪ/
dente /dɛnte/	*diente* /djente/	*tooth* /tuθ/
dito /dito/	*dedo* /dedo/	*finger* /fɪŋgər/
dare /dare/	*dar* /dar/	*give* /gɪv/

- There is no **SYSTEMATIC** correspondence #d : #d : #d here
- In fact, *día* and *day* are **FALSE COGNATES** (randomly similar)
- And, in fact, *diente* and *tooth* **ARE** cognates, even though they are very dissimilar

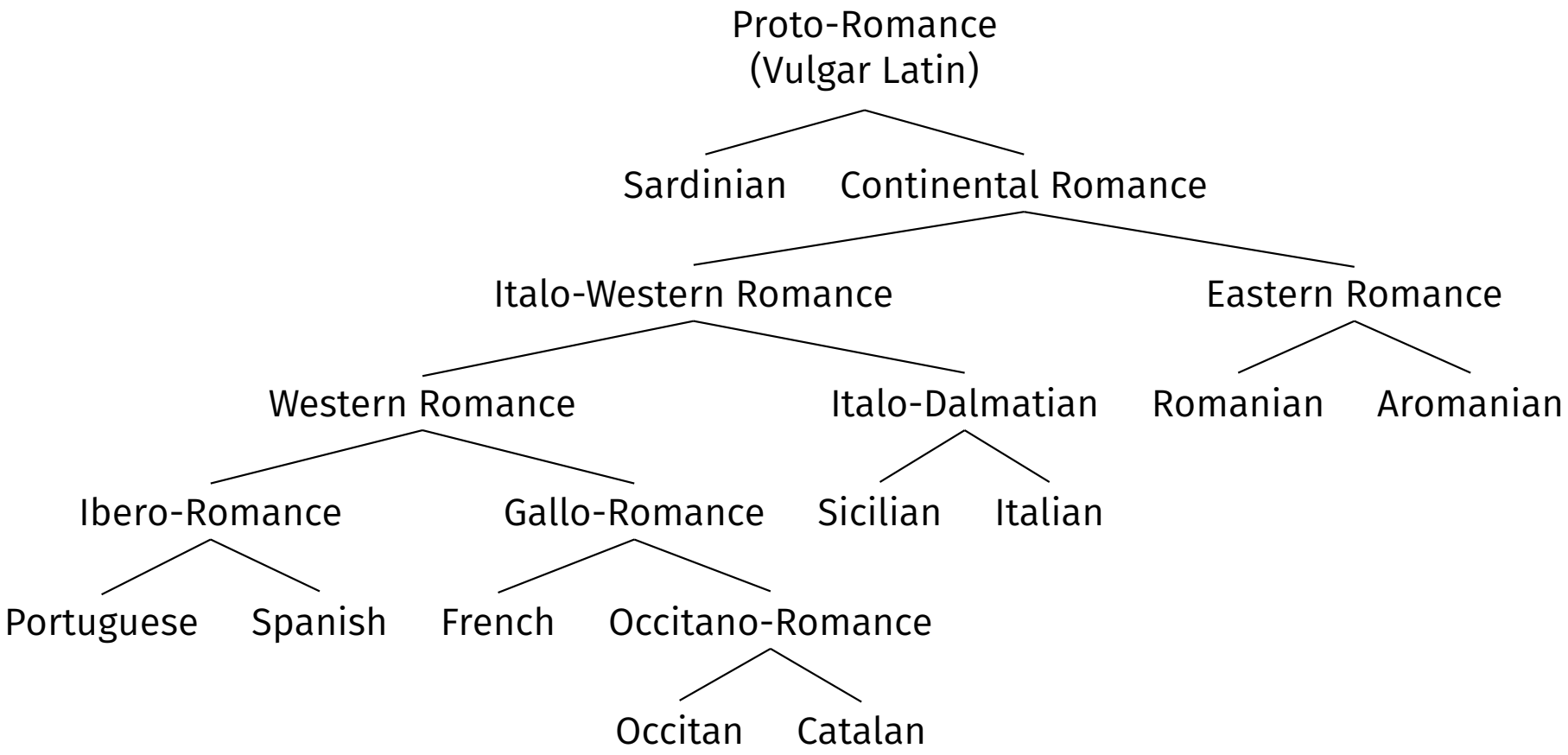
⇒ English is not a Romance language (but it is more remotely related to the Romance languages)

Subgrouping

- We've established that Italian, Spanish and French are all related and thus belong to the same family
- Can we establish any more detailed relationships?
- With more data, yes ⇒ Spanish and French form a **SUBGROUP**
- (Remember: both Spanish and French have undergone lenition between sonorants, while Italian hasn't)

Subgrouping

- With a lot more work, the Romance family tree takes something like the following form:



Exercise

- Now I want you to take a stab at working with the CM
- Working with the attached data from Quechua, reconstruct as many **CONSONANTS** of the protolanguage as you can
- Then try to find a subgrouping for the three languages
- Work with your neighbours! (Faster, more instructive and more fun.)
- If you run out of time, finish at home
- We'll discuss the exercise next week